NIKOLA

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE

NIKOLA

TRE FCEV

HYDROGEN FUEL CELL / ELECTRIC

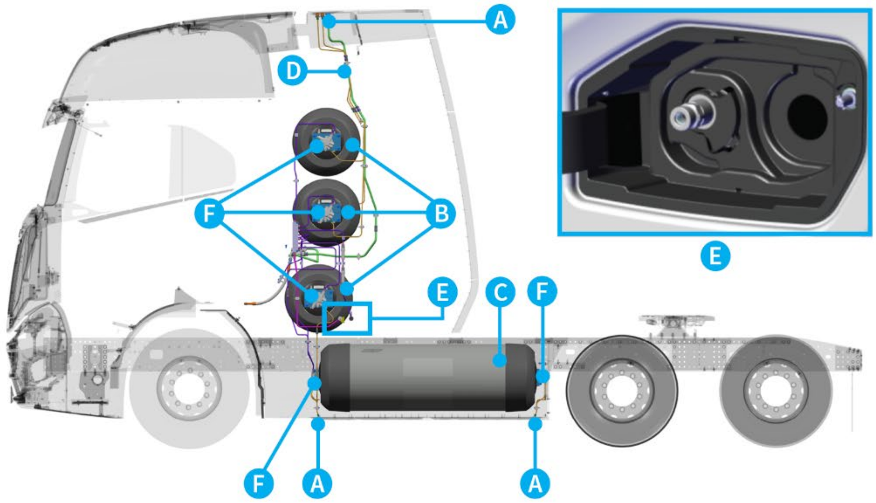
VERSION: 007

PEHS-MAN-TFCERG
Rev 7

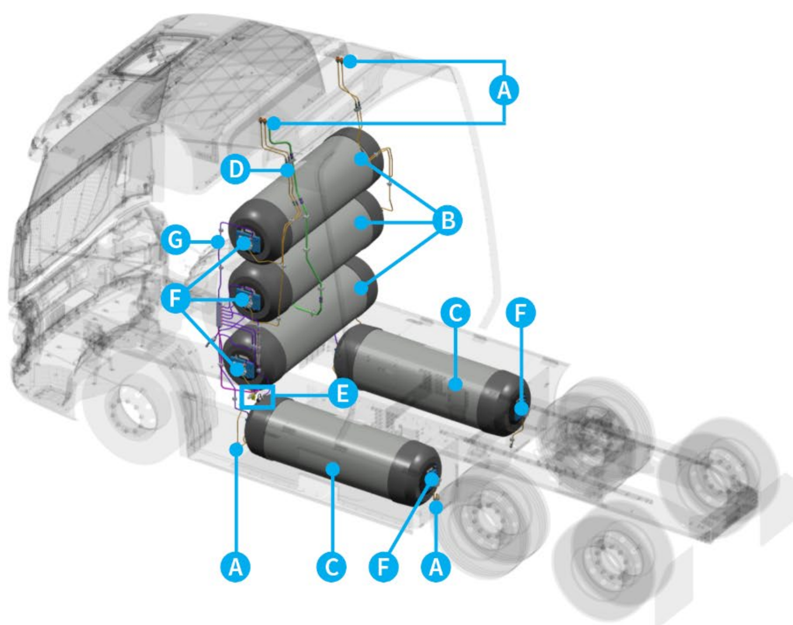
CONTENTS

Hydrogen Storage System (HSS) Diagram

DRIVER SIDE VIEW

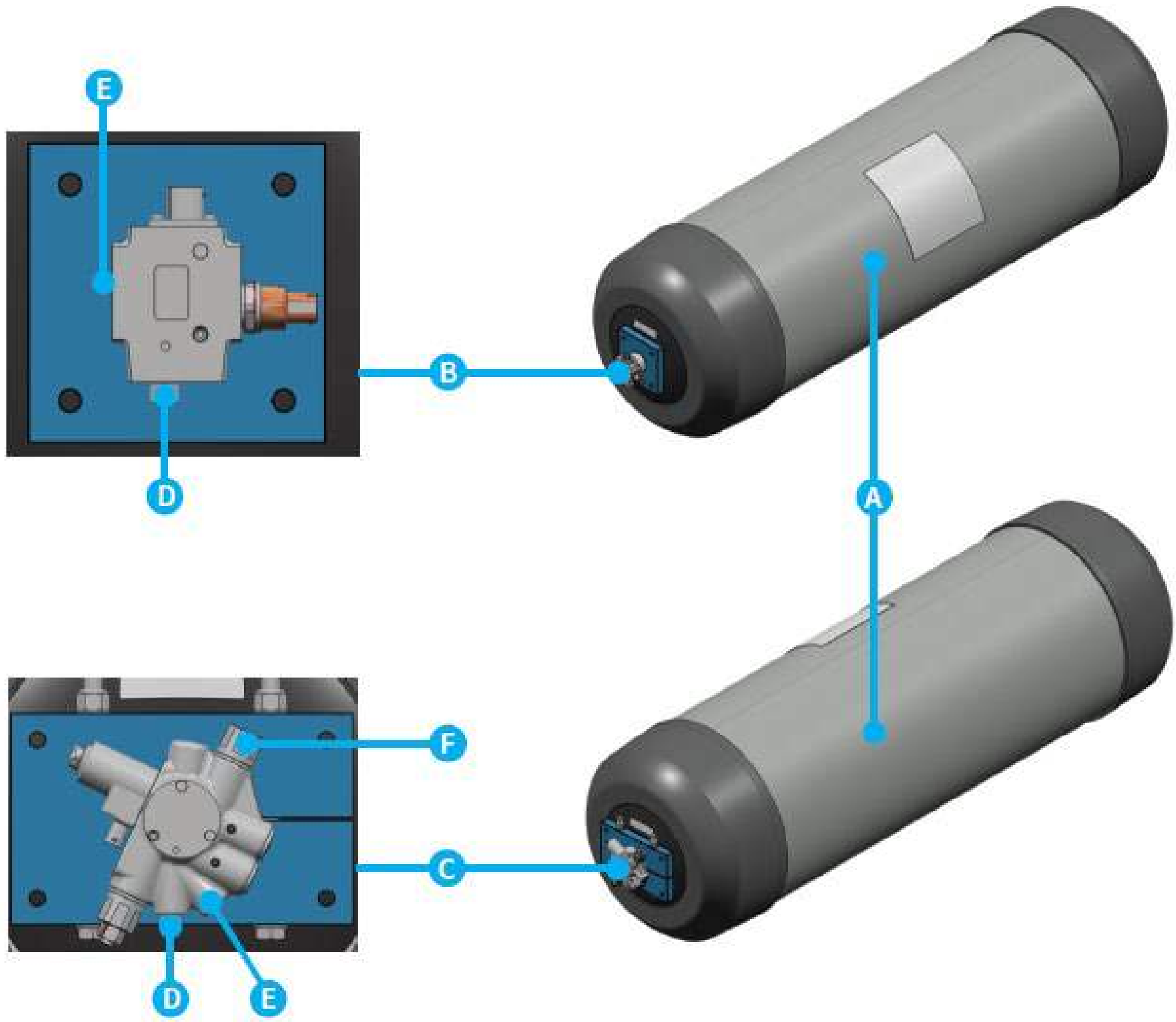


ISOMETRIC VIEW



- **A.** Thermally activated Pressure Relief Device (TPRD) Venting Outlets
- **B.** Backpack Hydrogen Storage Tanks (x3)
- **C.** Side Rail Hydrogen Storage Tanks (x2)
- **D.** Hydrogen Vent Lines
- **E.** Hydrogen Fueling Ports (H70)
- **F.** TPRD
- **G.** Hydrogen Fuel Lines

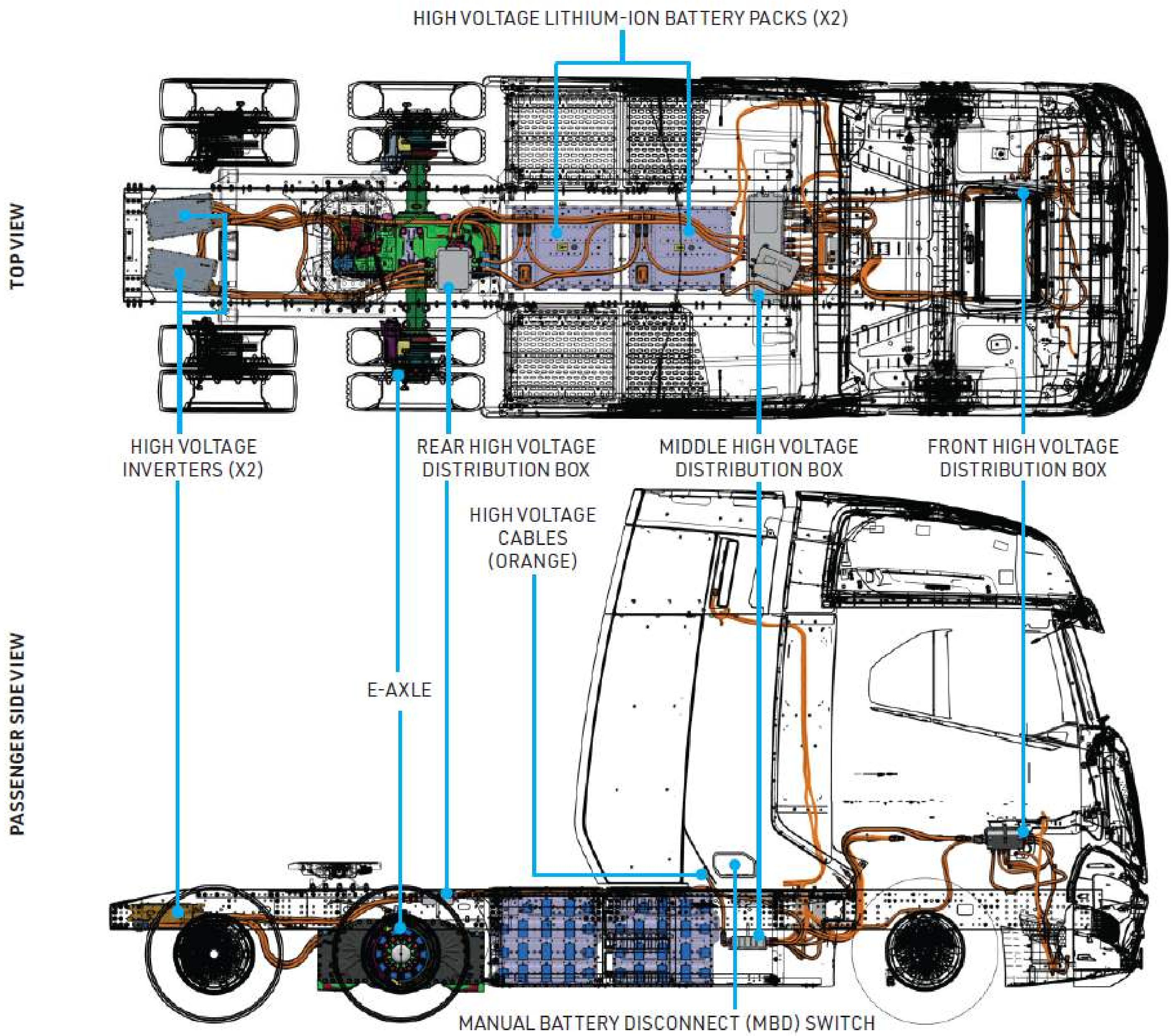
Hydrogen Storage Tank Diagram



A. Hydrogen Storage Tank
B. End Plug (includes TPRD)
C. On-Tank Valve (OTV)
D. TPRD
E. TPRD Vent
F. Fuel Line Inlet

High Voltage (HV) System Diagram

WARNING: All high voltage (HV) wires and harnesses are wrapped in orange insulation. NEVER cut these sections.
Do not try to touch without proper PPE (HV insulated gloves and boots, safety glasses, etc.).
Failure to follow these instructions may result in the risk of serious injury or death.



Hydrogen Safety

WARNING: Hydrogen is lighter than air and diffuses rapidly, therefore a small leak will quickly dissipate to a low concentration which cannot ignite.

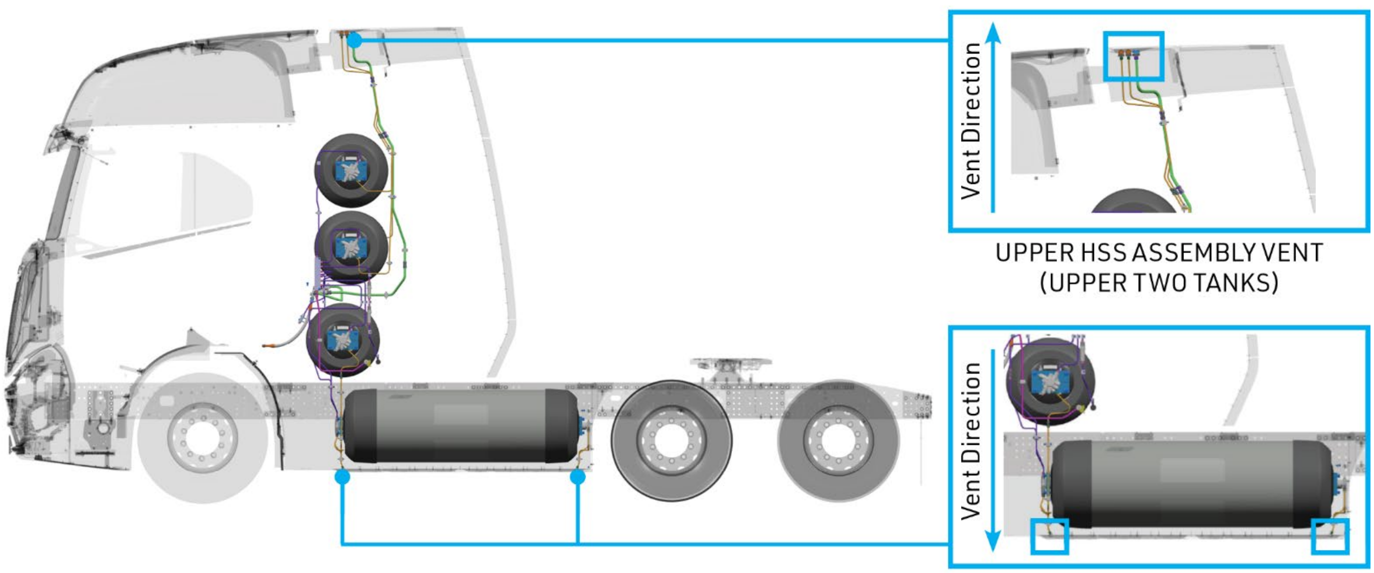
Hydrogen flames are not visible to the human eye. Hydrogen also burns very quickly and radiates significantly less heat than gasoline. If available, use a thermal imaging camera to detect hydrogen fires.

The flammability range of hydrogen is 4% to 74% (40,000 to 740,000 ppm) in air. Use a properly calibrated gas detection device to check hydrogen concentration near the vehicle.

Failure to follow these instructions may result in the risk of serious injury or death.

The Thermally activated Pressure Relief Devices (TPRD's) detect elevated temperatures and then vent the hydrogen tanks down to zero pressure in case of a vehicle fire. The TPRD will open at **230°F ± 9°F** (**110°C ± 5°C**) and discharge the stored hydrogen through vent-lines to specific areas of the vehicle per regulation.

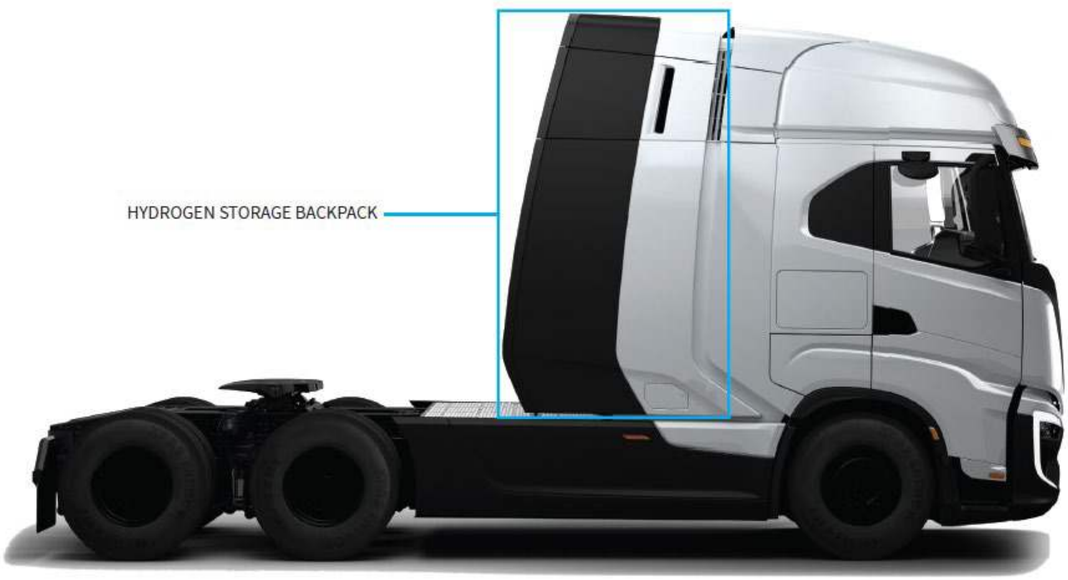
Evacuate the immediate area when the vehicle is on fire.



LOWER HSS ASSEMBLY VENT
(LOWER THREE TANKS)

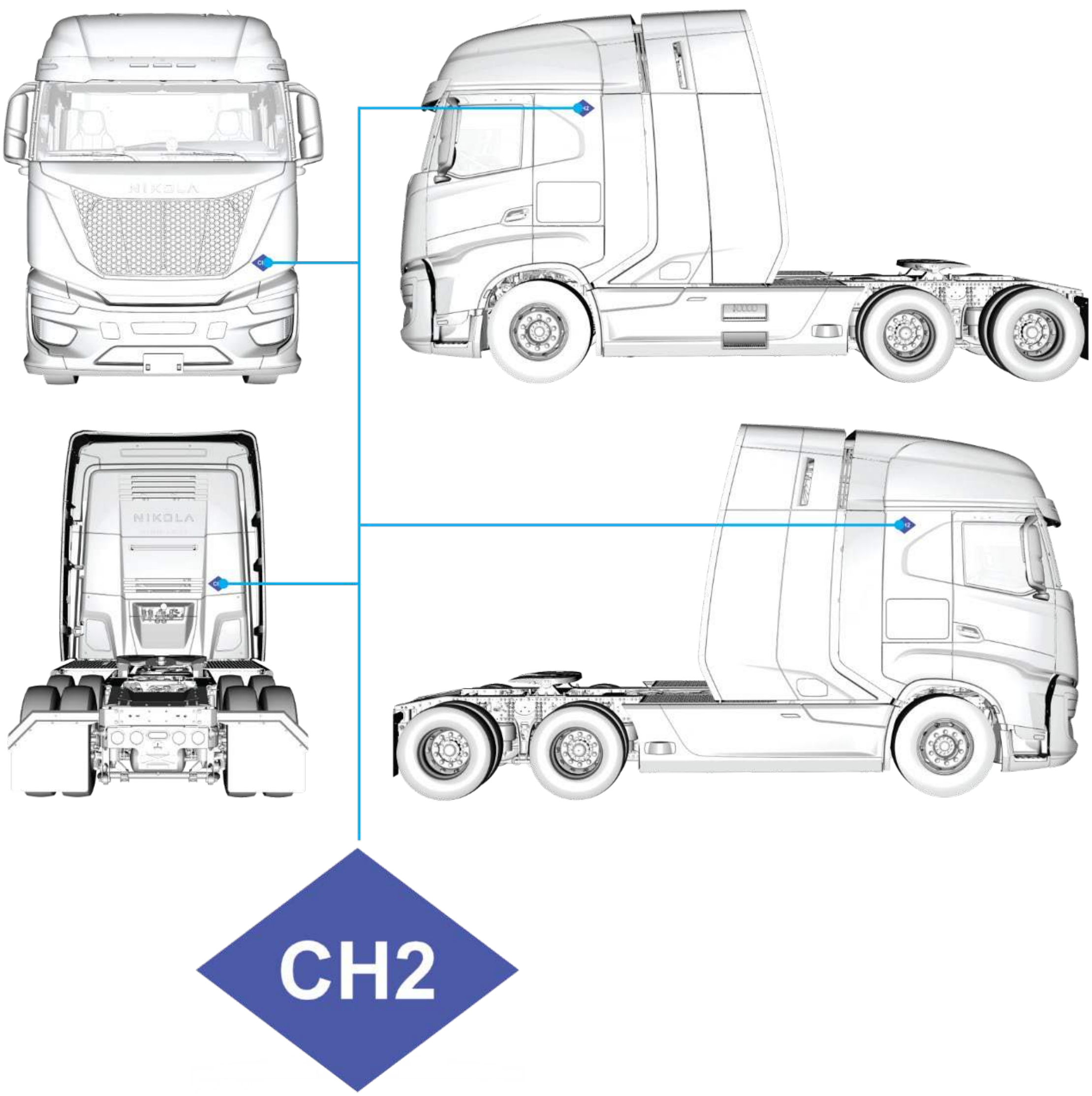
In case of hydrogen leakage, the vehicle has hydrogen detection sensors designed to automatically detect the leak and close all the on-tank valves (OTVs).

1. Identification / Recognition



CH2 Labels:

All FCEV vehicles will have CH2 labels on all four sides of the vehicle.



Vehicle Identification Number (VIN):

All Nikola manufactured vehicles are equipped with a HV system and can be identified as a Nikola vehicle by the first three characters of the VIN which will always be “1N9.”

Furthermore, the 6th character of the VIN will either be a "B" to indicate battery electric vehicle or an "H" to indicate hydrogen fuel cell electric vehicle.

The VIN is found on the driver side A-pillar and stamped on the frame railing inside driver side wheel well.

2. Immobilization / Stabilization / Lifting

WARNING: Do not touch, cut, or open high-voltage components and/or high-voltage batteries.

Do not cut or damage the Compressed Hydrogen Storage System, hydrogen pipes, or hydrogen-related components.

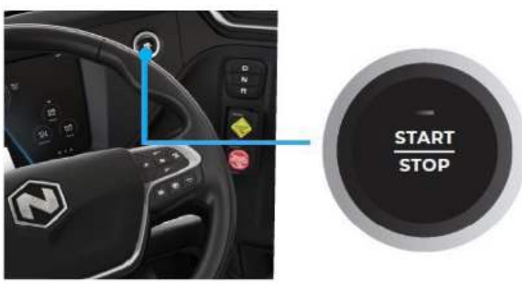
Always wear the full firefighting PPE and proper insulated electrical PPE (safety glasses, gloves rated for at least 1,000 V, HV insulated safety shoes, etc.).

Remove all metallic jewelry including rings, chains, and watches.

Failure to follow these instructions may result in the risk of serious injury or death.

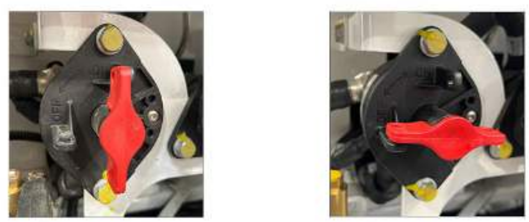
A. Chock the trailer and truck wheels.

B. Activate the parking brakes by lifting up on the right side of both the yellow and red switches.



C. If the vehicle is on, and time allows, turn it off by pressing the START | STOP button.

MBD ON MBD OFF

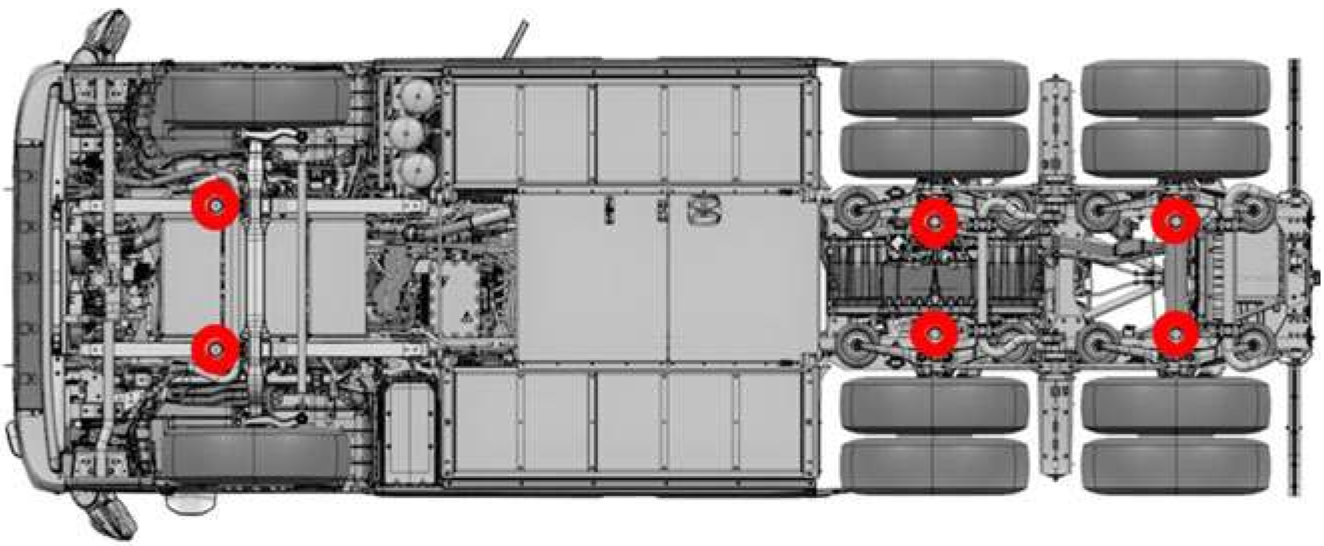


D. Turn the MBD switch counterclockwise (CCW) to off. See page 10.

Lifting Points:

The Vehicle can be lifted/jacked at the points along each axle shown in the figure below.

NOTICE: Never stabilize or lift the vehicle directly from the battery pack or anywhere along the high voltage system.



3. Disable Direct Hazards / Safety Regulations

Disabling High Voltage (HV):

WARNING: It is possible for the HV system to keep significant levels of voltage for a short duration after the system has been deactivated. Allow 5 minutes for voltage to de-energize before interacting directly with HV components.

Do not touch, cut, or open high-voltage components and/or high-voltage batteries.

Always wear full firefighting and proper insulated electrical PPE (safety glasses, gloves rated for at least 1,000V, HV insulated safety shoes, etc.). Remove all metallic jewelry including rings, chains, and watches.

Failure to follow these instructions may result in the risk of serious injury or death.

A. Locate the Manual Battery Disconnect (MBD) access door on the passenger side of the vehicle. Pull the panel door out and down from the top to open.

B. Locate the MBD switch.

C. Turn the MBD switch counterclockwise (CCW) to off.

D. Wait a minimum of 5 minutes for the high voltage system to de-energize.

MBD ON

MBD OFF

4. Access to the Occupants

Windows:

WARNING: Electrical and mechanical components (seats, doors, steering wheel, etc.) may be compromised after a collision.

If extrication is needed, do not touch, cut, or open high-voltage components and/or high-voltage batteries.

Failure to follow these instructions may result in the risk of serious injury or death.

The side windows and sunroof glass on the Tre FCEV are made of tempered glass.
The windshield is made of laminated safety glass.

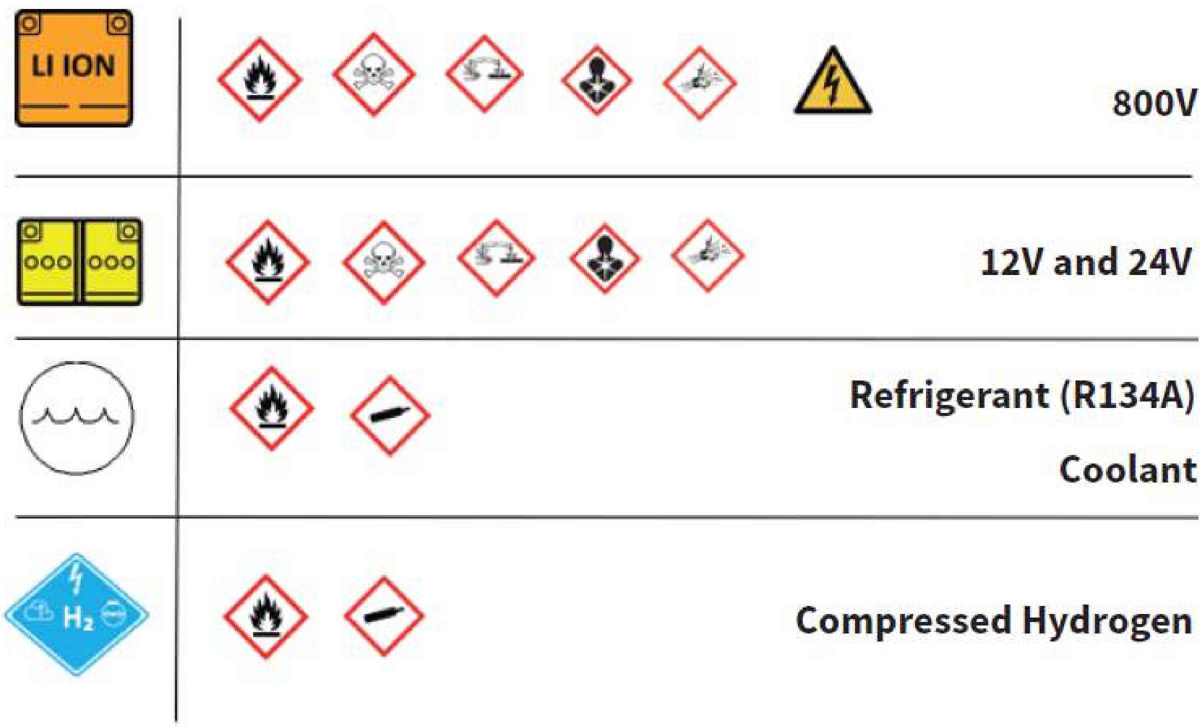
1. Tempered Glass
2. Laminated Safety Glass

5. Stored Energy / Liquids / Gases / Solids

WARNING: If coolant and/or refrigerant escapes from the system, there is a risk of thermal reaction within the high voltage battery system.

If available, use thermal imaging to monitor temperature of the high voltage battery for thermal reaction.

Failure to follow these instructions may result in the risk of serious injury or death.



6. In Case of Fire

If the fire is detected early enough when it is small and contained in a single area, spray enough water to quickly extinguish the flame.

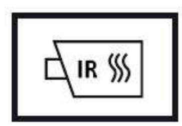
If fire has reached the Compressed Hydrogen Storage System, **DO NOT SPRAY WATER ON THE COMPRESSED HYDROGEN STORAGE SYSTEM.** Spraying water could potentially cool the Thermally-activated Pressure Relief Devices (TPRD) and prevent proper operation.

Do not store a vehicle holding a burned Li-Ion HV battery in or within 50 feet of a structure or other vehicle until the battery can be discharged.

> **WARNING:** If the hydrogen is on fire, extinguishing the hydrogen flame completely may cause unburned hydrogen to accumulate and lead to a secondary explosion.
> Follow all safety guidelines and procedures before entering the emergency scene.
> All personnel should wear and use full firefighting PPE and SCBA as needed at all vehicle fires.
> A battery fire may continue after hydrogen has dissipated. Use copious amount of water to extinguish.
> Failure to follow these instructions may result in the risk of serious injury or death.

USE COPIOUS AMOUNTS OF WATER TO FIGHT A HIGH VOLTAGE BATTERY FIRE.

If the battery catches fire, is exposed to high heat, or is generating heat or gases, use large amounts of water to cool the battery. It can take over 2,600 gallons (9,842 liters) of water (applied directly to the battery) to effectively cool and extinguish a high voltage battery fire.



If a Lithium Ion (Li-Ion) HV battery is involved in a fire, there is a possibility that it could reignite after being extinguished. If available, use thermal imaging to watch the battery. Re-apply water as necessary to cool the battery packs. Battery packs must be kept below 248°F (120°C). Alternatively, burying the battery packs in sufficient amounts of sand to cover the entire vehicle mid-section may achieve a similar effect and can be used to prevent re-ignition of battery packs.

Do not store a vehicle containing a damaged or burned Li-Ion HV battery in or within 50 feet of a structure or other vehicle until the battery can be discharged.

7. In Case of Submersion

DANGER: Damaged electric vehicles submerged in water present a potential high voltage electrical shock hazard. Exercise caution and wear proper PPE safety gear, including, but not limited to, high voltage safety gloves and boots.

Failure to follow these instructions will result in the risk of serious injury or death.

Fuel cell electric vehicles that have been submerged in water should be handled with greater caution due to the potential risk of a high voltage electrical battery fire. First responders should be prepared to respond to a potential fire risk. Pull the vehicle out of the water and allow the water to drain out of the vehicle's high voltage battery packs. Follow Section 3 of this document, ***Disable Direct Hazards / Safety Regulations***, once the vehicle is out of the water.

8. Towing / Transportation / Storage

Towing

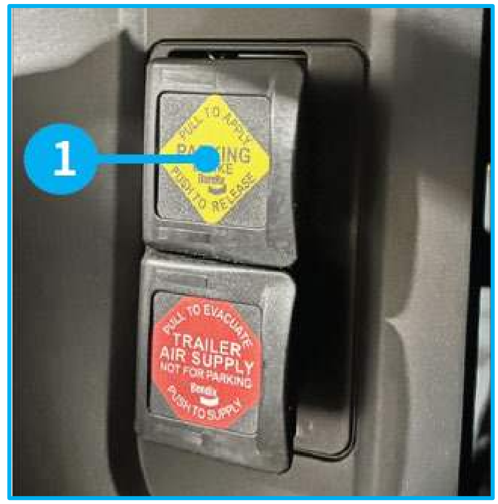
The vehicle can be pushed or pulled a short distance to clear it from the roadway. Once cleared, refer to the towing procedures to tow the vehicle. If you can, place the vehicle in Neutral, and disengage the parking brake prior to pushing or pulling.

NOTICE: Damage to the E-axle. Pulling the vehicle with the E-axle wheels turning may cause significant damage to powertrain components. When pulling the vehicle with the E-axle wheels turning, DO NOT exceed a speed of 3 mph (4.8 km/h) for a short distance. Pulling the vehicle in this manner should be extremely limited and only used for repositioning the vehicle to safely hook with a proper procedure.

The E-axle in the vehicle can generate power if the wheels spin during towing which could lead to overheating and cause significant damage. The vehicle can be towed from the rear (reverse tow or 5th wheel lift method), the front (with front hook), or with a lowboy trailer. Always transport the vehicle with the E-axle off the ground and restrict the wheels from spinning. The vehicle must remain in Neutral during towing.

Front Hook

1. Put the vehicle in the accessory mode.
2. Set the parking brake **(1)** and install wheel chocks along the tag axle wheels.

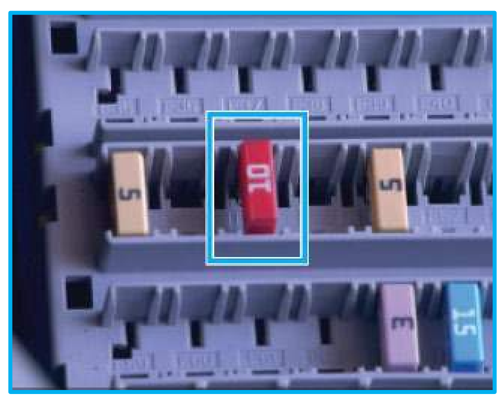


3. Locate the ECAS remote control on the left side of the driver's seat.

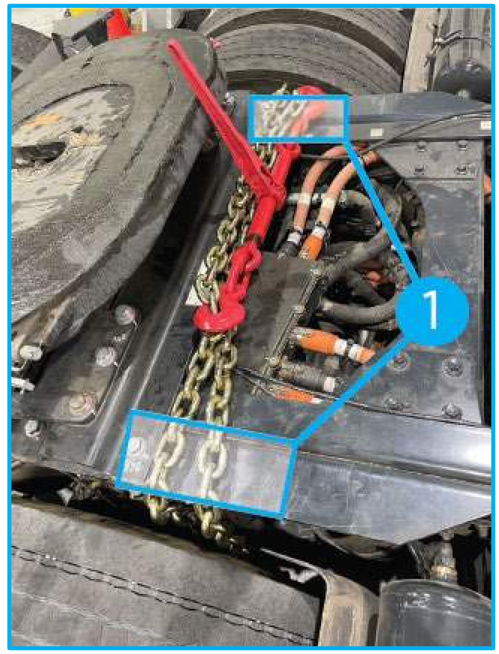
4. Press the consent button **(1)** on the ECAS remote control.
5. Press the lowering button **(2)** to set ECAS to the lowest position.

6. Deactivate the ECAS by switching off the vehicle power and removing 10A fuse from position **F49** in the passengers side dashboard fuse panel.

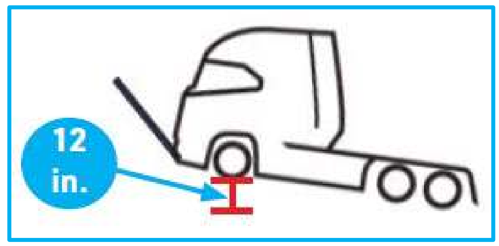
   Note: This will deactivate ECAS and will not compensate for any ride height changes during the towing procedure.



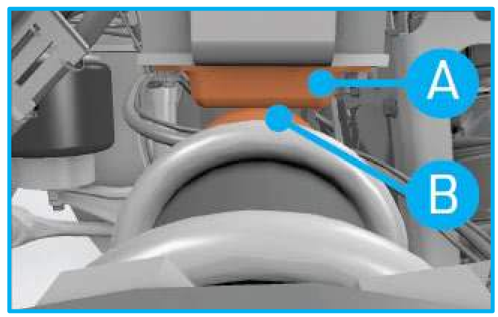
7. Chain the E-Axle to the frame.
8. Place two blocks **(1)** between the frame rails and chains to keep the chains from damaging the frame rails.



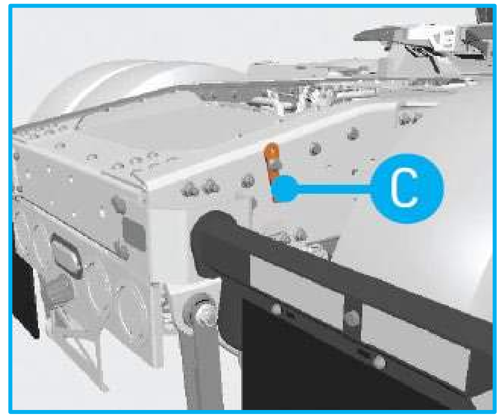
9. Lift the vehicle up from the front axle such that the tires are 12 inches above the ground.



Note: This tilts the truck and loads the tag axle more and compresses the tag axle air suspension. The tag axle **(B)** will contact the bump stops **(A)** if the vehicle's front end is lifted while the trailer is still hitched.

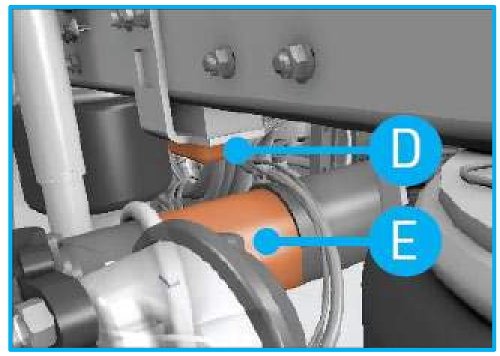


10. Locate the manual air inlet port **(C)** on the tag axle.



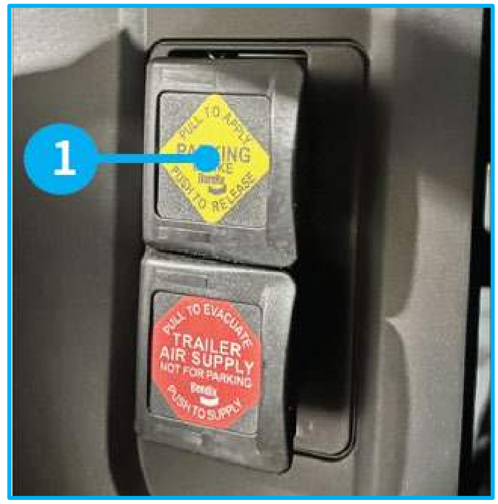
11. Use the manual air inlet port and an external air source to pressurize the tag axle air springs to lift the frame above the tag axle so that there is **2 in. (52 mm)** of clearance between the bottom of the rubber portion of the bump stop **(D)** and to the top of the tag axle **(E)**.

    Note: This allows for tag axle jounce travel while towing with the trailer.

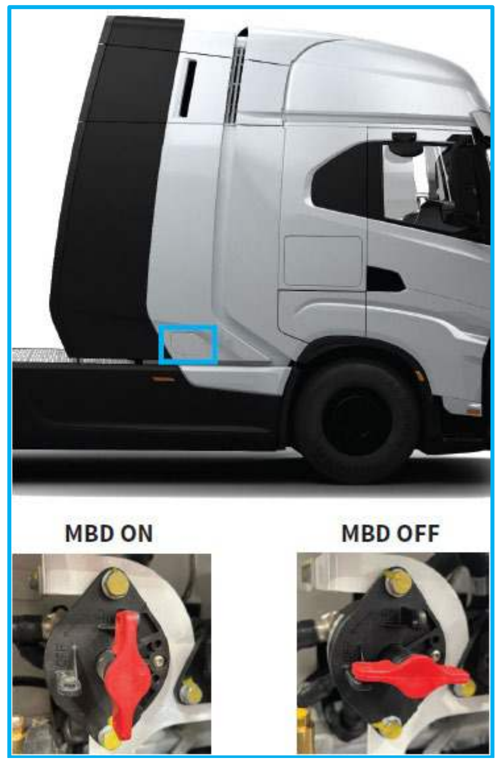


12. Disconnect the external air source line from the manual air inlet port.

13. Disengage the parking brake **(1)**.



14. Turn vehicle off in the cab.
15. Set the MBD switch to the OFF position.



16. Remove the wheel chocks from the tag axle wheels.

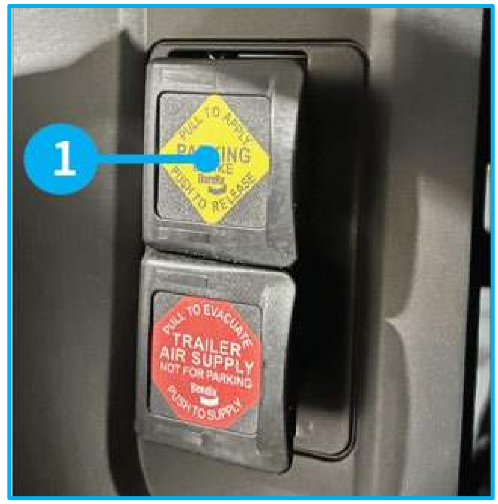
    Note: The vehicle is now ready for towing.

17. Place all tow kit contents back into the tow kit package, including the previously removed ECAS fuse.

PROCEDURE IS COMPLETE

Rear Hook (no trailer) with Reverse Kingpin / 5th Wheel Lift

1. Put the vehicle in the accessory mode.
2. Set the parking brake **(1)** and install wheel chocks along the front axle wheels.



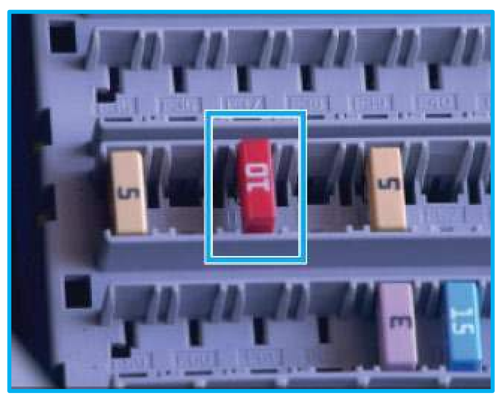
3. Secure the steering wheel into a non-moving position.

4. Locate the ECAS remote control on the left side of the driver's seat.

5. Press the consent button **(1)** on the ECAS remote control.
6. Press the lowering button **(2)** to set ECAS to the lowest position.

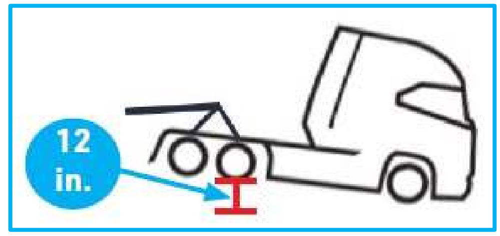
7. Deactivate the ECAS by switching off the vehicle power and removing 10A fuse from position **F49** in the passengers side dashboard fuse panel.

   Note: This will deactivate ECAS and will not compensate for any ride height changes during the towing procedure.

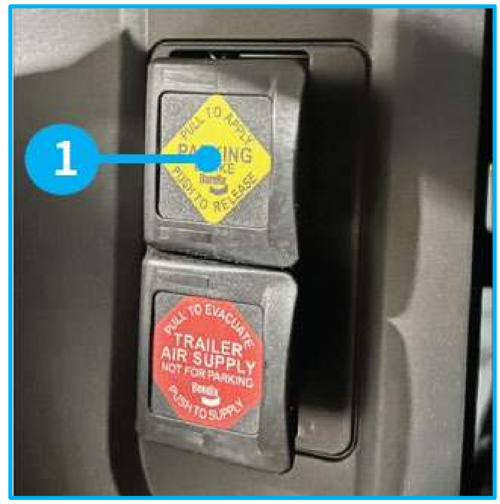


8. Chain the E-axle and tag axle in a triangle formation, avoiding contact with the air tanks.

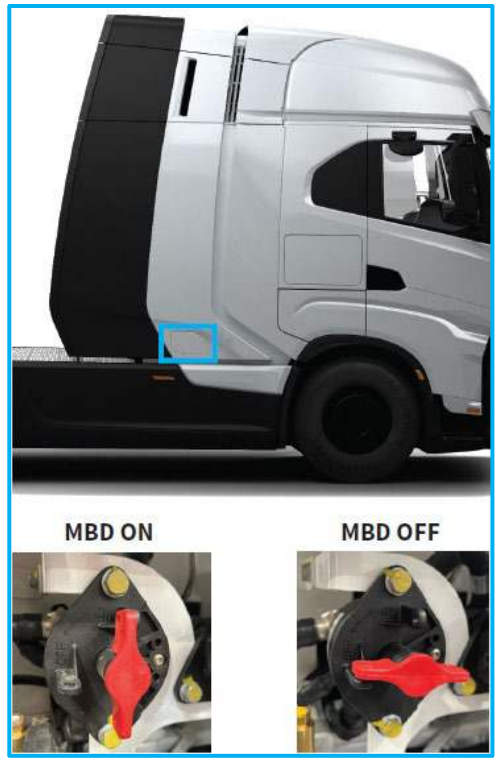
9. Lift the vehicle up from the rear axles such that the tires are 12 inches above the ground.



10. Disengage the parking brake **(1)**.



11. Turn vehicle off in the cab.
12. Set the MBD switch to the OFF position.



13. Remove the wheel chocks from the front axle wheels.

    Note: The vehicle is now ready for towing.

14. Place all tow kit contents back into the tow kit package, including the previously removed ECAS fuse.

PROCEDURE IS COMPLETE

Transportation

High voltage and hydrogen fuel systems may be compromised because of a collision. Before transporting, it is important to assume these components are energized. Always follow high voltage and hydrogen systems safety precautions and avoid transporting until the HV system has been properly deenergized. Failure to do so may result in serious injury.

Storage

WARNING: The potential for battery re-ignition exists after a crash or fire incident. Vehicle should be stored outside at a safe distance (50 ft minimum) from other vehicles and structures. Failure to follow these instructions may result in the risk of serious injury or death.

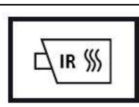
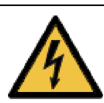
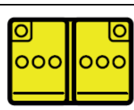
Damaged fuel cell electric vehicles should be kept in an open area (instead of in a garage or other type of enclosed storage facility) until the batteries have been properly discharged.

For further instructions on storing and/or scrapping fuel cell and high voltage battery system components, contact Nikola Pulse Uptime Center at **+1 (888) 268-1181**.

9. Important Additional Information

First responders and training officers with questions may contact Nikola Pulse Uptime Center at **+1 (888) 268-1181**.

10. Explanation Pictograms Used

	Acute Toxicity
	Battery Electric Vehicle
	Notice / Caution / Warning / Danger
	Corrosive
	Explosive
	Flammable
	Fuel Cell Electric Vehicle
	Gases under pressure
	Hazards to the Human Health
	High Voltage Battery
	Infrared Imaging
	High Voltage Warning
	Low Voltage Battery